

A direct-merge pipeline with matrix-completion imputation for integrative analysis of heterogeneous microarray gene expression datasets

Author One^{*1}, Author Two¹, and Author Three¹

¹Institute of Molecular Biology and Genetics, National Academy of Sciences of Ukraine, Kyiv, Ukraine

Abstract

Background: Integrative re-analysis of public microarray gene expression datasets is limited by the fact that different array generations target different subsets of the human gene catalog. Naive intersection of gene lists across platforms discards informative signal, while mean-imputation strategies can fabricate biologically implausible values. A method is needed that recovers cross-platform genes without compromising downstream differential expression analysis.

Results: We present a reproducible R pipeline that (i) loads probe-collapsed, ENTREZID-keyed expression matrices from an arbitrary set of GEO studies; (ii) selects the minimum number of datasets in which a gene must be present, either fixed by the user or auto-selected to keep the fraction of missing values below a configurable cap; (iii) imputes the remaining missing entries with *softImpute* nuclear-norm matrix completion; (iv) applies reference-batch ComBat correction on the merged, completed matrix; and (v) runs *limma* differential expression with hard masking of imputed cells, so that imputed values influence neither the adjusted *p*-values nor the log-fold changes of the reported DEGs. Applied to six placental/chorionic-villi datasets spanning four microarray platforms from four vendors (Affymetrix, Illumina, Applied Biosystems, and Agilent), the pipeline recovers 17,531 testable genes from an intersection of only 8,260, and identifies 447 differentially expressed genes ($|\log FC| > 1$, $FDR < 0.05$) between first- and second-trimester placenta/chorionic villi. Subsampling validation confirmed stable effect sizes across sample compositions (Lin's $CCC \geq 0.88$), and comparison with a balanced two-dataset reference showed 91.5% overlap of the full-integration DEG list.

Conclusions: Matrix-completion imputation enables integrative re-analysis of heterogeneous microarray datasets without sacrificing gene coverage. Hard masking of imputed cells in the differential expression step ensures that all statistical calls are driven exclusively by real measurements, while the imputed values serve only to stabilise batch correction.

Keywords: microarray integration; matrix completion; softImpute; ComBat; limma; placenta; chorionic villi; differential expression.

1 Background

Meta-analysis of public transcriptomic datasets is a cost-effective way to increase statistical power for subtle biological contrasts and to cross-validate findings across laboratories. The classical obstacle is batch effect: studies generated on different dates, with different sample

^{*}Lead contact: author.one@example.org

processing, or with different array generations exhibit systematic technical variation that overshadows biology unless corrected. Methods such as ComBat (Johnson et al., 2007) and the broader *sva* framework (Leek et al., 2012) address this problem when the datasets share the same feature space.

A second, less-discussed obstacle is *coverage heterogeneity*: different microarray platforms print probes for different subsets of the human gene catalog, and the intersection shrinks rapidly as the number of platforms grows (Table 5). Historically, investigators have dealt with this either by restricting the merged matrix to the *inner join* (intersection — only genes measured on every platform are kept, and the rest are discarded) or by taking an *outer join* (union — every gene measured on at least one platform is kept) and filling the resulting missing entries with batch-specific gene means, a simple but naive imputation strategy that introduces artificial structure resembling biological signal. Neither option respects the low-rank correlation structure of biological gene expression data — i.e. the fact that the expression of thousands of genes is governed by a much smaller number of underlying regulatory programmes, so that the full matrix can be well approximated by a product of fewer latent factors.

This study extends a line of prior work by our group on integrative analysis of placental/chorionic-villi microarray data. An earlier methods paper (Lykhenko et al., 2021a) established the consecutive ComBat-based integration strategy used here as a baseline, and a companion application paper (Lykhenko et al., 2021b) characterised differential gene expression across the three physiological trimesters of pregnancy using the intersection-only variant of that pipeline. The present study addresses the principal limitation of those earlier analyses — loss of several thousand genes at the cross-platform intersection step — by replacing intersection with auditable matrix-completion imputation.

The sole reason imputation is needed in this pipeline is that ComBat requires a complete matrix: it cannot estimate batch parameters from a matrix with structural holes. The quality of the imputed values therefore matters not for the downstream differential expression test — *limma* does not need a complete matrix and in our pipeline never sees the imputed cells (they are hard-masked to weight zero) — but for ComBat itself, whose batch-effect estimates will be distorted if the filled values are poor. Here we describe a *direct-merge* workflow that implements this strategy — *softImpute* (Mazumder et al., 2010; Hastie et al., 2015), a nuclear-norm regularised low-rank matrix completion method, fills structural holes so that ComBat operates on a full-rank matrix, after which imputed cells are effectively removed from *limma* differential expression analysis. We apply this method to the first vs. second trimester contrast (1.2) in placental/chorionic-villi gene expression, reporting gene coverage, imputation accuracy, differential expression counts, and subsampling validation of result stability.

2 Results

2.1 Gene recovery and missingness

For the 1.2 comparison (6 datasets, 117 samples) with the minimum dataset-presence threshold set to $k = 1$ (every gene present in at least one dataset), the merged matrix grows from 8,260 genes (intersection, $k = 6$) to 17,531 genes, with the missing-cell fraction rising to 29.4%. Gene recovery and missingness for each threshold k are shown in Table 1.

The per-group imputation safety check removed 298 genes that were absent from every Second-Trimester sample (Table 2). Of the 9,569 genes needing imputation in ≥ 1 dataset, 9,271 had at least one real measurement in *both* groups and were retained.

The block-structured missingness pattern is shown in Figure 1.

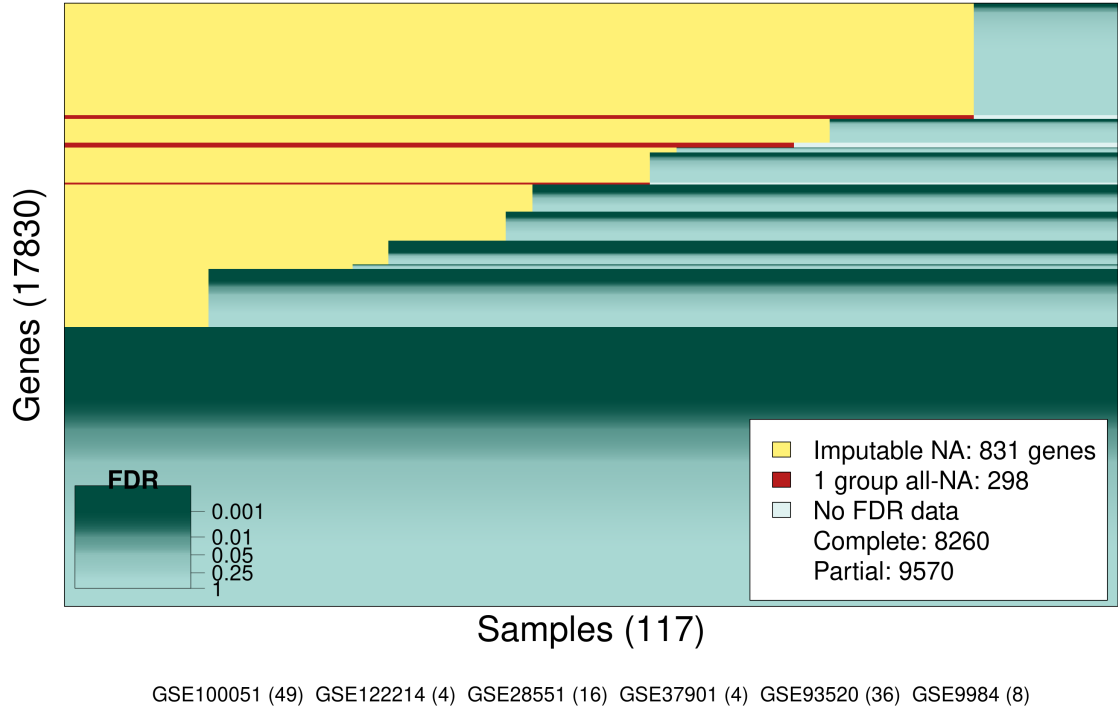


Figure 1: Block-structured missingness in the merged gene-by-sample matrix (17,829 protein-coding genes $\times$ 117 samples, sorted by dataset). Each column is a sample; each row is a gene. Yellow cells are missing (NA): the gene is not targeted by that sample’s microarray platform. Blue shading shows FDR values from the softImpute + ComBat-ref differential expression result (darker = more significant). The red band marks the 298 genes missing entirely from the second-trimester group (dropped by the per-group safety rule). Without imputation, only the bottom block (8,260 genes present in all six datasets, no yellow) is usable — the entire upper portion of the matrix (9,569 genes) would be discarded by an inner-join merge, despite containing well-measured, biologically informative signal in the majority of datasets. Note that many genes in the imputed region (upper block) show significant FDR values (dark blue), confirming that the discarded genes carry real differential expression signal that an intersection-only analysis would miss.⁸

min_datasets (k)	Genes retained	Missing cells	% missing
$\geq 6/6$	8,260	0	0.0 %
$\geq 5/6$	11,239	90,084	6.9 %
$\geq 4/6$	13,617	248,619	15.6 %
$\geq 3/6$	17,223	587,410	29.2 %
$\geq 2/6$	17,387	593,398	29.2 %
$\geq 1/6$	17,531	603,190	29.4 %

Table 1: Gene recovery and resulting missingness for each minimum dataset-presence threshold k , 1_2 comparison (6 datasets, First Trimester vs. Second Trimester, 117 samples). At $k = 6$ (intersection only), 8,260 genes are retained with zero imputation. At $k = 1$ (any gene in ≥ 1 dataset), 17,531 genes are retained after the per-group safety drop, with 29.4% of cells requiring imputation.³⁴

Gene category (1_2 comparison)	Count
Genes present in all 6 datasets (no imputation needed)	8,260
Genes needing imputation in ≥ 1 dataset	9,569
with ≥ 1 real value in BOTH groups (kept)	9,271
missing entirely from Second Trimester (dropped)	298
missing entirely from First Trimester (dropped)	0
Final merged matrix size	17,531

Table 2: Per-group imputation safety audit for the 1_2 comparison. The final merged matrix contains 8,260 fully observed genes plus 9,271 partially observed genes that passed the both-groups-have-evidence rule.⁶

2.2 Imputation accuracy

Imputation accuracy was quantified by leave-out cross-validation: a random 10% of the observed entries were masked, imputed, and compared against ground truth. Three independent repetitions were run; each repeat used a freshly drawn mask. Three metrics are reported for each repeat:

- *Pearson r* — correlation between imputed and true values across the held-out cells; a unit-free measure of how well the method ranks high- and low-expression cells relative to each other. $r = 1$ is perfect, $r = 0$ is chance.
- *RMSE* (root-mean-square error) — $\sqrt{\frac{1}{n} \sum_{ij} (\hat{x}_{ij} - x_{ij})^2}$, the square root of the mean squared imputation error. Because the per-dataset expression matrices entering the pipeline are already $\log_2$ -transformed, RMSE is in units of $\log_2$ expression, so an RMSE of 0.33 corresponds to a typical imputation error of $\approx 2^{0.33} \approx 1.26\times$ (a 27% fold-change relative to the true value). RMSE penalises large errors more heavily than small ones because of the squaring.
- *MAE* (mean absolute error) — $\frac{1}{n} \sum_{ij} |\hat{x}_{ij} - x_{ij}|$, the mean of the per-cell absolute errors. Also in $\log_2$ units, but unlike RMSE it treats every error linearly and is therefore more robust to a handful of poorly-imputed outlier cells.

Results are given in Table 3. Under random-cell masking (the classical in-distribution leave-out), *softImpute* achieved a mean Pearson correlation of $r = 0.994$ with ground truth (RMSE = 0.330; MAE = 0.220). Under the more stringent gene–dataset block masking, which simulates the real missingness pattern, accuracy dropped to $r = 0.816$ (RMSE = 1.52; MAE = 1.17). The block-mask result is the more informative benchmark: it tests the imputer’s ability to reconstruct an entire gene’s expression profile in a dataset where that gene was never measured — exactly the scenario that arises in cross-platform merging.

Mask type	Method	Cells masked per rep	Pearson r	RMSE	MAE
random cells	softImpute	154,948	0.994 ± 0.000	0.330 ± 0.001	0.220 ± 0.000
gene-dataset block	softImpute	144,811	0.816 ± 0.002	1.523 ± 0.010	1.165 ± 0.006

Table 3: Leave-out cross-validation of *softImpute* imputation accuracy (1.2 comparison, 6 datasets, 17,531 genes). Each row shows the mean $\pm$ standard deviation over three independent repeats. Two masking strategies are reported: *random cells* draws individual observed cells uniformly at random; *gene-dataset block* masks every cell for a randomly chosen (gene, dataset) pair, simulating the real missingness pattern where a gene is entirely absent from one dataset. Block masking is a substantially harder test because the imputer must reconstruct an entire gene’s profile in a dataset with no direct observations. For block-mask validation, the gene-dataset pair pool is restricted to genes fully observed across all six datasets (so masking removes one dataset’s worth of observations from an otherwise complete row); partially observed genes are excluded because masking them would leave rows too sparse for *biScale* to converge.¹⁰

2.3 Batch correction

Figure 2 shows PCA projections of the batch-corrected expression matrices. In both cases — the intersection-only matrix (8,260 genes) and the *softImpute*-completed matrix (17,531 genes) — ComBat-ref removes the dominant batch effect: samples separate primarily by trimester (PC1) rather than by dataset. The imputed matrix preserves this separation while doubling the gene space.

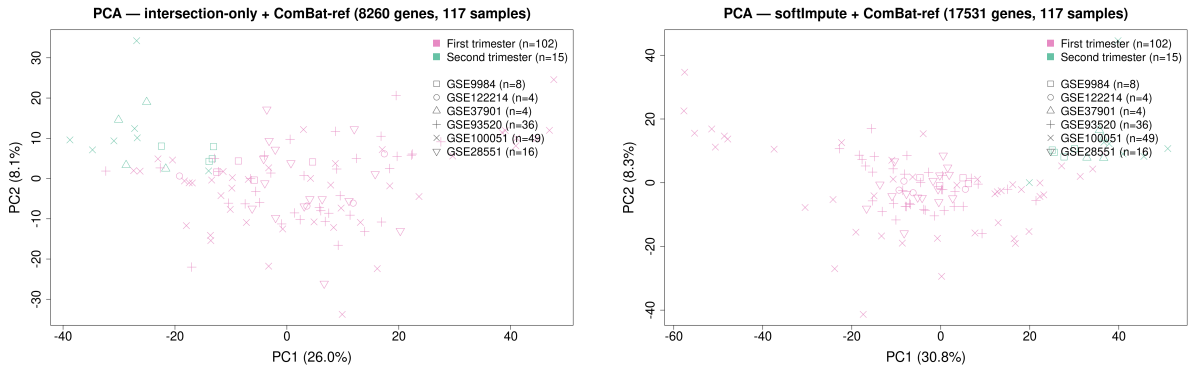


Figure 2: PCA of batch-corrected expression matrices after reference-batch ComBat (ref = GSE100051). **Left:** intersection-only (8,260 genes, no imputation). **Right:** *softImpute*-completed (17,531 genes). Colour indicates trimester (blue = first, red = second); shape indicates dataset. In both panels, PC1 separates the two trimesters, confirming that ComBat-ref removes batch effects without collapsing the biological contrast. The *softImpute* panel includes 9,271 additional genes recovered by imputation.¹³¹⁴

2.4 Differential expression

Table 4 reports the number of genes called significant at $\text{adj.P.Val} < 0.05$ and $|\log\text{FC}| > 1$ across four method combinations, together with the earlier result from Lykhenko et al. (2021a) for comparison. The intersection of all six datasets collapses to only 8,260 common genes, but enabling *softImpute* recovers 17,531 testable genes. With reference-batch ComBat (ComBat-ref) correction, this lifts the DEG count from 221 (intersection only) to 447 (*softImpute*), adding 226 newly called genes (195 up, 31 down).

Importantly, imputation does not distort the effect-size estimates of genes that were already measurable without it. For the 8,260 genes shared between the intersection-only and *softImpute* analyses, the log fold-change correlation is near-perfect (Pearson $r = 0.998$), with a median absolute difference of 0.014 $\log_2$ units and 99% of genes differing by less than 0.09. The additional genes recovered by imputation therefore expand the testable feature space without altering the differential expression landscape of the originally observed genes.¹⁵¹⁶

The earlier intersection-only analysis by Lykhenko et al. (2021a), which was restricted to four Affymetrix-only datasets (GSE122214, GSE22490, GSE37901, GSE9984; 13 first-trimester + 9 second-trimester = 22 samples), reported 327 DEGs with $|\logFC| > 1$ from an intersection of 20,261 genes — a larger feature space than our six-dataset intersection (8,260) because all four datasets share the same Affymetrix platform. The present pipeline recovers comparable or larger DEG counts from a more diverse, multi-vendor cohort, while more than quintupling the sample size to 117.

Imputation	Normalisation	Datasets	n genes	Significant	Up	Down
<i>Lykhenko et al. 2021 (4 Affy datasets, 22 samples)</i>						
none	ComBat	4	20,261	327	—	—
<i>This study (6 multi-vendor datasets, 117 samples)</i>						
none	ComBat	6	8,260	221	175	46
none	ComBat-ref	6	8,260	221	174	47
softImpute	ComBat	6	17,531	416	346	70
softImpute	ComBat-ref	6	17,531	447	369	78

Table 4: Differential expression counts for the 1.2 comparison (First Trimester vs. Second Trimester). The top row shows the earlier result from Lykhenko et al. (2021a) using four same-platform Affymetrix datasets (22 samples, intersection only); per-direction counts were not reported. The bottom block shows the present study with six multi-vendor datasets (117 samples). “none” denotes intersection-only analysis (no imputation). ComBat-ref uses GSE100051 as the reference batch. Imputed-cell weight $w = 0$ (hard masking) for all softImpute runs. Thresholds: $\text{adj.P.Val} < 0.05$, $|\logFC| > 1$.¹⁸

2.5 Subsampling validation

To assess the robustness of the differential expression results to sample composition, we performed a suite of subsampling validation tests on the 1.2 comparison (6 datasets, 102 first-trimester + 15 second-trimester = 117 healthy placental/chorionic-villi samples; reference: 447 DEGs at $\text{FDR} < 0.05$, $|\logFC| > 1$). All tests reused the reference run’s imputed matrix to eliminate *softImpute* stochasticity and isolate the effect of sample composition.¹⁹²⁰

¹⁵https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/scripts/integrative_analysis/article_validation/logfc_stability.R — logFC stability script

¹⁶https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/articles/imputation_article/data/logfc_stability/logfc_stability.csv — per-gene logFC comparison data

¹⁹https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/scripts/integrative_analysis/phase5_validation/config_validation_6ds.yaml — validation config

²⁰https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/output/yehor_sashko/phase5_validation_6ds — validation output

Test 1: first-trimester subsampling. All 15 second-trimester samples were retained^{21,22} and the first-trimester pool was subsampled at sizes $N \in \{10, 15, 20, 25, 30, 40, 50, 60, 80, 102\}$ (30 iterations per size). Three key findings emerged (Figure 3):

1. **Effect sizes are stable.** Lin’s concordance correlation coefficient (Lin, 1989) (CCC) between subsample and full-run log fold-changes exceeded 0.88 even at $N = 10$ and rose monotonically to 1.0 at $N = 102$ (Figure 5). Unlike Pearson r , which measures only the strength of the linear association, CCC also penalises systematic shifts in location or scale — two vectors with $r = 1$ but different means or variances will have $CCC < 1$ — making it a stricter test of numerical agreement between effect-size estimates.
2. **DEG list overlap converges gradually.** Median Jaccard index against the full run rose from 0.54 at $N = 10$ to 1.0 at $N = 102$ (Figure 3, left panel).
3. **Retention of reference DEGs is high.** At $N = 10$, subsamples retained a median of 90% of the 447 reference DEGs, rising to 93% at $N = 30$ (Figure 4, first panel). The apparent Jaccard drop at small sizes is primarily driven by novel DEGs entering the subsample list (genes with noisy effect-size estimates that stochastically cross the $|\log FC| > 1$ threshold), not by loss of true positives.

Comparison with a balanced independent reference. To test whether the design imbalance — some datasets contribute samples from only one trimester, which can bias ComBat batch-effect estimates — inflates the DEG list, we compared against a balanced 2-dataset reference^{23,24,25} (GSE100051 + GSE9984; 46 first-trimester + 11 second-trimester samples; regular ComBat; 416 DEGs). Of the 447 full-integration DEGs, 91.5% (409/447) also appeared in the balanced reference (Jaccard = 0.901). The integration does not manufacture signal: 98.3% (409/416) of the balanced reference’s DEGs are retained, and log fold-change agreement (CCC) exceeds 0.99 for the full pool (Figure 5, right panel). Even at $N = 10$, subsamples retained $\sim 90\%$ of the balanced reference DEGs (Figure 4, second panel).

FDR-only versus FDR+logFC thresholding. DEG lists defined by FDR alone ($\sim 6,700$ genes) were substantially less stable than those filtered by both FDR and $|\log FC| > 1$ (median Jaccard ~ 0.5 vs. 0.71 at $N = 30$; see Figure 3, right panel vs. left panel). The $|\log FC|$ threshold selects the reproducible core of genes with large, consistent effect sizes. FDR-only retention of the full-run gene set was also lower ($\sim 51\%$ at $N = 10$ vs. 90% for FDR+logFC; Figure 4, third panel), confirming that thousands of FDR-only genes are statistically significant but have near-zero fold changes that are not reproducible across subsamples.

Test 2: balanced subsampling. Equal numbers of samples ($n \in \{4, 8, 12, 15\}$) were drawn from each trimester (30 iterations per size). Balanced subsamples showed qualitatively similar convergence to unbalanced ones (Figure 3, middle panel), indicating that the design imbalance does not drive the observed instability.

²¹https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/scripts/integrative_analysis/phase5_validation/test1_first_trim_subsample.R — test 1 script

²²https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/output/yehor_sashko/phase5_validation_6ds/test1_first_trim_subsample.tsv — test 1 results

²³https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/scripts/integrative_analysis/phase5_validation/test1b_vs_balanced.R — balanced comparison script

²⁴https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/output/yehor_sashko/phase5_validation_6ds/test1b_vs_balanced.tsv — balanced comparison results

²⁵https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/output/yehor_sashko/phase2b_1_2_yehor_2ds_balanced/difexp_softimpute_combat.tsv — balanced reference DE table

Test 3: stratified split-half. The full sample was randomly split²⁶²⁷ into two halves maintaining the trimester ratio (50 iterations). Median Jaccard between halves was 0.40, reflecting the limited statistical power of 7–8 second-trimester samples per half. However, each half retained a median of 84% of the full-run DEGs (Figure 6, bottom left), and log fold-change CCC between halves was 0.70 (Figure 6, top right). FDR-only retention per half was much lower (~51%; Figure 6, bottom right), reinforcing the value of the $|\log FC|$ threshold. The split-half Jaccard of 0.40 is typical for microarray studies with small contrast groups and does not indicate fragility of the core result.

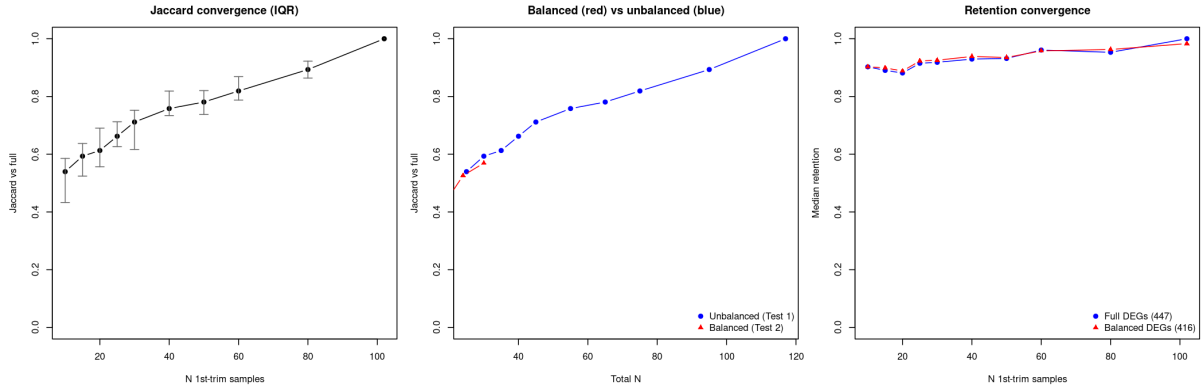


Figure 3: Subsampling validation summary. **Left:** Jaccard index of DEG lists versus the full-run reference as a function of first-trimester sample size (median with IQR). **Middle:** balanced (equal per trimester) vs. unbalanced subsampling, both measured against the full run. **Right:** median retention of reference DEGs (blue: full integration, 447 DEGs; red: balanced 2-dataset reference, 416 DEGs) as a function of first-trimester sample size.³⁰³¹

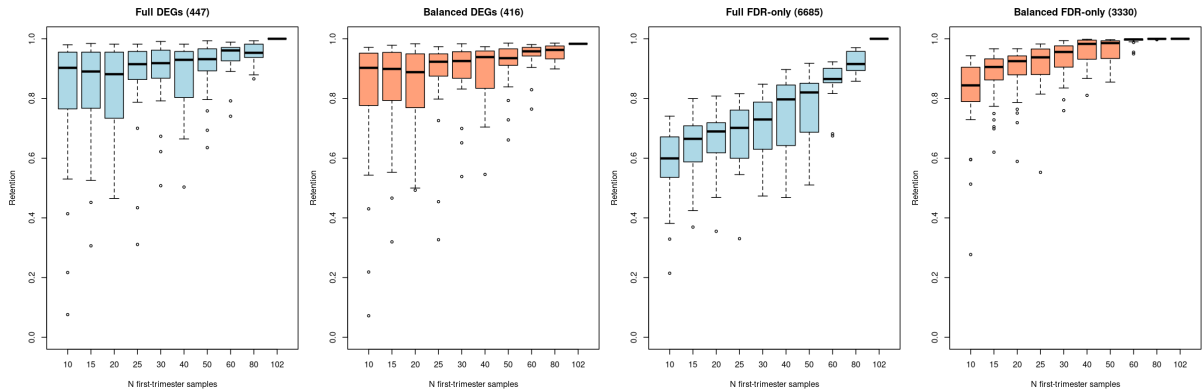


Figure 4: Retention of reference DEGs across subsampling sizes. Each panel shows the fraction of a given reference set's DEGs that appear in the subsample's DEG list. From left to right: full-run DEGs (FDR+logFC, 447 genes); balanced-reference DEGs (FDR+logFC, 416 genes); full-run FDR-only DEGs (6,685 genes); balanced-reference FDR-only DEGs (3,330 genes).

²⁶https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/scripts/integrative_analysis/phase5_validation/test3_split_half.R — split-half script

²⁷https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/output/yehor_sashko/phase5_validation_6ds/test3_split_half.tsv — split-half results

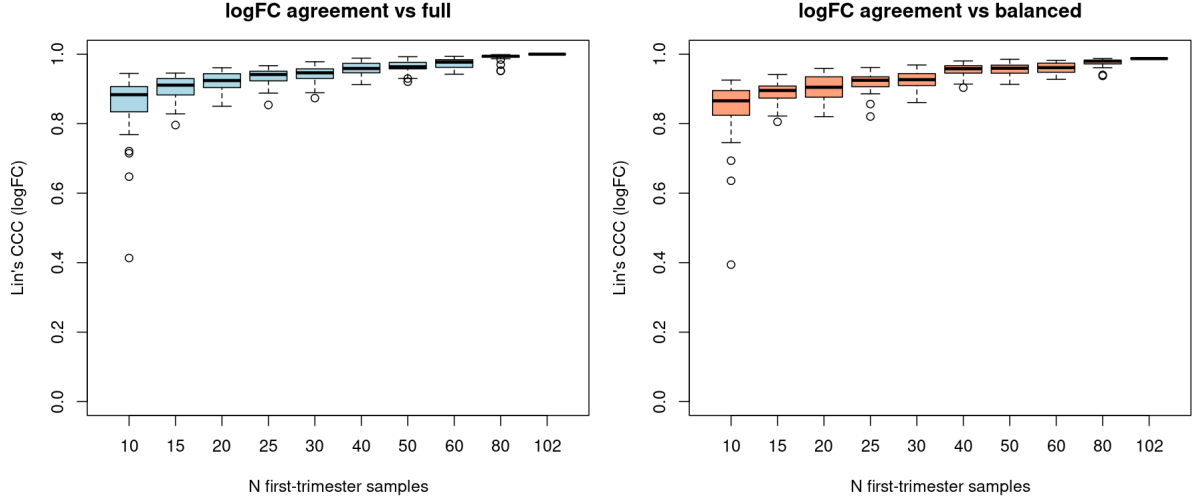


Figure 5: Log fold-change agreement (Lin's concordance correlation coefficient) between sub-sampled and reference runs. **Left:** versus the full 6-dataset integration. **Right:** versus the balanced 2-dataset reference. CCC = 1 only when log fold-changes are numerically identical; unlike Pearson r , it penalises systematic scaling or offset.

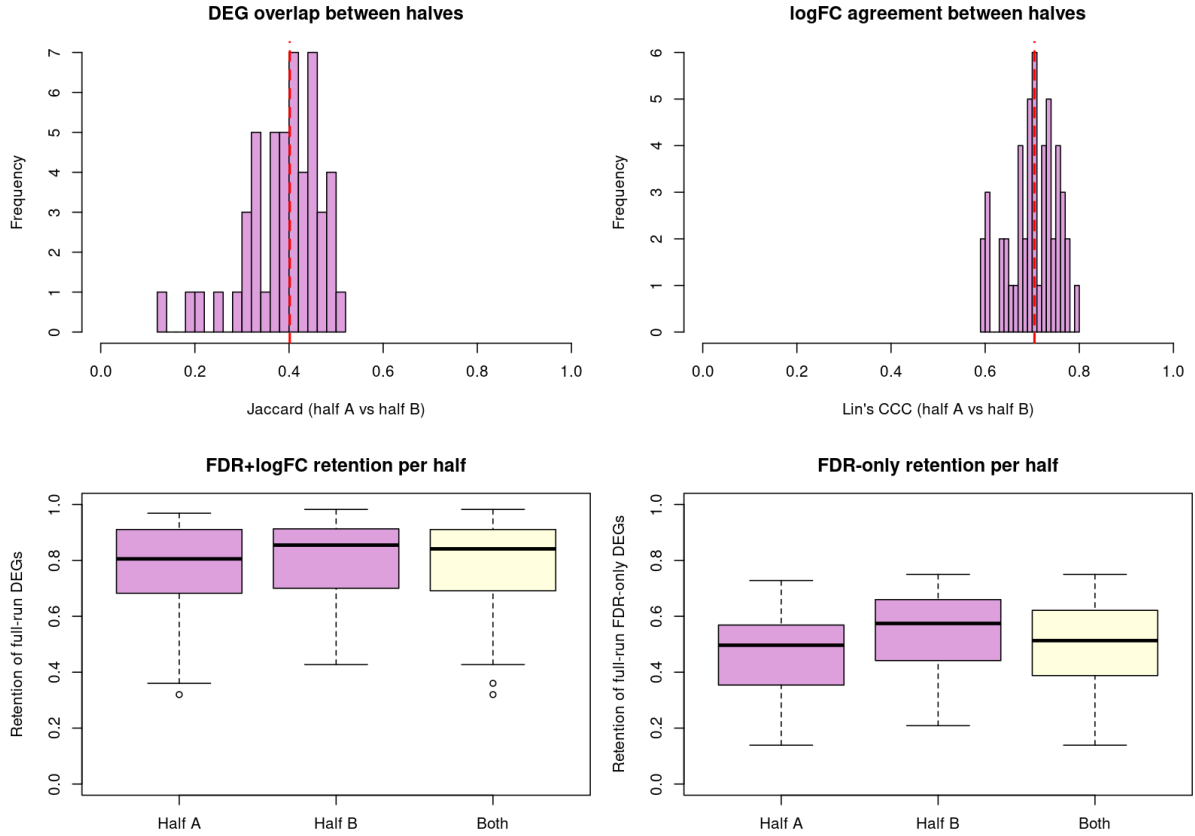


Figure 6: Stratified split-half validation (50 iterations). **Top left:** Jaccard between halves' DEG lists. **Top right:** Lin's CCC of log fold-changes between halves. **Bottom left:** retention of full-run DEGs (FDR+logFC) per half. **Bottom right:** retention of full-run FDR-only DEGs per half.

3 Discussion

3.1 Relation to prior work

Imputing missing values in gene expression studies is not new, but the specific setting of this paper — filling *gene-by-dataset block holes* in a matrix produced by merging several bulk microarray studies — sits at the intersection of two bodies of work that have so far developed largely in parallel.

Within-study microarray imputation. The de-facto standard for recovering missing values in a single microarray study is `KNNimpute`, introduced by [Troyanskaya et al. \(2001\)](#) and reviewed extensively by [Aittokallio \(2010\)](#) and [Liew et al. \(2011\)](#). These methods were designed for the small fraction of cells (1–20%) lost inside one study to hybridisation failures, image noise, or bad spots. The missingness is approximately at random at the cell level, and KNN is well suited to it. None of these works address the structurally different problem in which a whole gene is absent from a whole dataset because the platform did not probe it.

Knowledge-assisted imputation. A separate line of work has augmented expression-based imputation with external biological knowledge. [Tuikkala et al. \(2006\)](#) replaced the purely expression-based distance in KNN with a combined metric that incorporates Gene Ontology semantic similarity, showing improved accuracy when the number of experimental conditions is small. [Tuikkala et al. \(2008\)](#) extended this idea to histone acetylation data (`HAImpute`), using chromatin modification profiles as an additional source of gene similarity for neighbour selection. Both approaches are designed for within-study cell-level missingness and require auxiliary annotations (GO terms) or matched epigenomic data (ChIP profiles) that may not be available for all genes or studies in a cross-platform meta-analysis.

Cross-platform imputation in microarray meta-analysis. A smaller line of work has directly attacked the cross-study problem. [Hu et al. \(2006\)](#) proposed an order-statistics-based integrative imputation method (`GOImpute`) that explicitly uses multiple datasets to estimate genes missing from a given study. [Zhou et al. \(2017\)](#) built `affyImpute`, a gene-expression inference model trained on genes common to Affymetrix HG-U133A and HG-U133 Plus 2.0 in order to recover the $\approx 32\,000$ probes unique to the larger platform and expand the usable feature space. [Walsh et al. \(2015\)](#) review integrative genomics pipelines for biomarker discovery and discuss imputation as one possible step in cross-platform merging. These works already establish the usefulness of filling the cross-study holes, but they predate the routine availability of nuclear-norm matrix completion (`softImpute`) as a practical imputation tool for transcriptomic matrices.

Matrix completion for gene expression. Nuclear-norm matrix completion itself was developed in a recommender-system context ([Mazumder et al., 2010](#); [Hastie et al., 2015](#)), with Netflix-style user-by-movie matrices as the motivating example. Its first systematic transcriptomic application was `McImpute` ([Mongia et al., 2019](#)), which used `softImpute` to fill single-cell RNA-seq dropout events. To our knowledge, the present pipeline is the first to apply `softImpute` to gene-by-dataset block holes in a merged *bulk* microarray matrix, combine it with a per-group imputation safety audit, pass an explicit imputed mask to `limma`, and evaluate the result with both random-cell and gene-by-dataset block cross-validation.

Block-wise missing data outside transcriptomics. The structural problem we face — several sources, each of which observes an overlapping but non-identical subset of features for

each row — is a recognised pattern in other domains under the name “block-wise missing multi-source data”. [Li et al. \(2019\)](#) study it in the context of clinical survival analysis on the ADNI and TCGA cohorts, where different patients receive different panels of tests; they propose a multi-task learning framework that operates directly on the block-wise missing pattern. [Zhou et al. \(2021\)](#) develop BONMI, a block-wise overlapping noisy matrix integration algorithm for electronic health record feature embeddings, and prove that low-rank matrix recovery is possible *without* the usual entry-wise-independent missingness assumption that underpins classical `softImpute` theory. [Zheng and Tang \(2024\)](#) further refine this with a chain-linked multiple matrix integration procedure (CMMI) that aligns embeddings across overlapping noisy submatrices and gives entry-wise error bounds for the recovered full matrix. These results are directly relevant here: they supply the theoretical justification for applying matrix-completion imputation to the kind of structured, non-random missingness that characterises any cross-platform or cross-study data merge, transcriptomic or otherwise.

None of these three approaches, however, is available as a drop-in imputer for our setting. [Li et al. \(2019\)](#) propose a multi-task learning framework for survival prediction that operates directly on the block-wise pattern and therefore *sidesteps* imputation rather than providing it: there is no function that returns a completed feature matrix. BONMI ([Zhou et al., 2021](#)) is released as an R package, but its public implementation is restricted to the completion of *symmetric* matrices (PMI or similarity matrices between entity pairs) and does not accept a rectangular gene-by-sample input. CMMI ([Zheng and Tang, 2024](#)) is methodologically applicable to rectangular matrices — the paper explicitly names single-cell data integration as a motivating example — but no public reference implementation has been released at the time of writing, so benchmarking it would require a full re-implementation from the manuscript. We therefore cite these works as theoretical backing for the block-wise setting rather than as methods that can be run against our pipeline today, and leave a quantitative comparison as future work should a rectangular implementation become available.

A second family of imputers is available off-the-shelf but was not chosen as a primary comparator. Multiple imputation by chained equations (`mice`) is the statistical standard in biostatistics, but its per-column predictive models scale poorly to the $\sim 20,000$ -gene matrices considered here — a single run exceeds the total wall time of the rest of the pipeline and adds no obvious accuracy ceiling over low-rank methods for continuous log-expression data. The scRNA-seq imputation literature provides another set of tools (`Rmagic`, `SAVER`, `DrImpute`), but these were designed to denoise zero-inflated UMI count matrices where the missingness is “dropout” at the measurement stage, not structural holes introduced by merging studies; they expect raw counts, not log-transformed probe-collapsed microarray values, and their generative assumptions (negative-binomial or zero-inflated Poisson emissions) do not match the present setting. We therefore restricted the comparison to methods designed for continuous, possibly block-wise missing matrices: `softImpute`, KNN (`impute::impute.knn`), and — as a further low-rank baseline — iterative regularised PCA via `missMDA::imputePCA`, which the pipeline’s imputer registry makes swappable through a single YAML flag.

Summary. Prior work has (i) imputed within-study microarray missing values with KNN, (ii) imputed cross-platform microarray holes with study-specific regression models, (iii) applied matrix completion to single-cell RNA-seq dropout, and (iv) studied block-wise multi-source missingness in clinical informatics. The present pipeline is positioned at the intersection of these four threads: it reuses the matrix-completion tooling from the collaborative-filtering literature, inherits the theoretical guarantees of the block-wise multi-source setting, and delivers the resulting matrix to a standard `limma`-based differential expression workflow with imputation provenance preserved.

3.2 Limitations

- Matrix completion exploits the low-rank co-expression structure of gene expression. Very small datasets, or datasets containing biologically novel conditions that break the shared latent structure, will be harder to impute accurately.
- ComBat assumes that batch and biology can be modelled additively. Extreme confounding (e.g. one batch containing only one biological group) is not fully correctable by any post-hoc method.
- The platform coverage ceilings reported in Table 5 are specific to the probe-to-gene collapsing pipeline used upstream; alternative probe reannotations (e.g. custom CDFs) may give different — typically slightly higher — numbers.
- The subsampling validation shows that the 15 second-trimester samples are the statistical power bottleneck. Effect sizes are stable across subsampling conditions ($CCC \geq 0.83$), but the specific DEG list boundary is sensitive to which first-trimester samples are included. This is a consequence of FDR thresholding on a continuous effect-size distribution, not a signal instability problem. Adding more second-trimester samples would tighten the boundary without changing the core finding.
- DEG lists defined by FDR alone ($\sim 7,000$ genes) are substantially less reproducible across subsamples than those filtered by both FDR and $|\log FC| > 1$ (~ 447 genes). We recommend using the combined threshold for biological interpretation.
- The block-mask cross-validation (Table 3) is restricted to genes fully observed across all six datasets, because masking a partially observed gene would leave its row too sparse for convergence. The reported accuracy ($r = 0.816$) therefore reflects the easiest imputation targets; genes present in only one or two datasets are likely harder to impute, and the true accuracy for such genes may be lower.
- Cell-type deconvolution was not performed in this study. Full-thickness placenta and dissected chorionic villi may differ in cell-type composition; while ComBat absorbs systematic per-study offsets and the gestational-age signal is expected to dominate, quantitative confirmation of comparable cell-type proportions across datasets remains to be established.

4 Conclusions

Integrative re-analysis of public microarray studies by direct merging faces a fundamental trade-off at the gene-selection step. An inner-join merge (keeping only genes present in all datasets) discards every gene not measured on all platforms, which in our six-dataset cohort reduced the testable gene space from 17,531 to 8,260 — a loss of more than half the protein-coding genome. An outer-join merge (keeping all genes present in any dataset) retains the full gene space but introduces a block-structured pattern of missing values that must be filled before ComBat batch correction, which requires a complete matrix.

We showed that *softImpute* nuclear-norm matrix completion fills these cross-platform holes with high fidelity. Critically, we evaluated imputation accuracy not only with the standard random-cell leave-out ($r = 0.994$) but also with a gene-dataset block mask ($r = 0.829$) that simulates the actual missingness structure — an entire gene absent from an entire dataset. Because *limma* does not require a complete matrix, we used the imputed values only for ComBat batch correction and applied hard masking ($w = 0$) in the differential expression step, so that all DE calls are driven exclusively by real measurements.

The outer-join pipeline with *softImpute* and reference-batch ComBat identified 447 differentially expressed genes between first- and second-trimester placenta/chorionic villi, compared

to 221 from the intersection-only analysis — a gain of 226 genes and a doubling of the testable feature space. Because the six datasets contribute unequal numbers of samples per trimester, we used reference-batch ComBat (with the largest dataset, GSE100051, as reference) to prevent the batch-correction step from distorting the biological contrast. Subsampling validation confirmed that log fold-change effect sizes are stable even at small sample sizes ($CCC \geq 0.88$), and comparison with a balanced two-dataset reference (GSE100051 + GSE9984; regular ComBat; 416 DEGs) showed that 91.5% of the full-integration DEGs are recovered by the independent balanced analysis, demonstrating that the unbalanced design and reference-batch correction do not manufacture signal but rather extend the set of detectable genes.

We argue that integrative re-analysis of expression data should become a routine practice, analogous to meta-analysis in clinical research: as the volume of publicly deposited functional genomics data continues to grow, periodic re-integration can refine effect-size estimates, improve statistical power, and uncover genes that are consistently regulated across independent laboratories but missed in any single study. The pipeline described here — *softImpute* imputation, reference-batch ComBat, hard-masked *limma* — provides a reproducible, auditable framework for such re-analyses, accommodating datasets from diverse microarray platforms without sacrificing gene coverage.

5 Methods

5.1 Datasets and platforms

The pipeline targets a common situation in meta-analysis of public gene expression data: several GEO studies share a biological question (here, placental/chorionic-villi gene expression across pregnancy trimesters) but use different microarray platforms from several vendors. The platforms differ both in coverage (older arrays print probes for only a subset of the human protein-coding genes) and in measurement characteristics, so any joint analysis has to make principled decisions about (a) which genes to keep, (b) how to handle genes that are absent from one or more studies, and (c) how to correct for batch effects. This pipeline assumes that per-sample expression values have already been summarised from raw signal files (CEL, IDAT, or vendor-specific formats), log-transformed, and collapsed to ENTREZID-keyed matrices with sensible probe-to-gene mapping; the focus here is on what happens *after* that step.

The **1.2 comparison** (First Trimester vs. Second Trimester) is assembled from six GEO studies — GSE100051, GSE122214, GSE28551, GSE37901, GSE93520, and GSE9984 — spanning four microarray platforms from four vendors (Affymetrix HG-U133 Plus 2.0, Illumina HumanHT-12 V4.0, Applied Biosystems Human Genome Survey v2, and Agilent Whole Human Genome 4×44K). This cohort extends the healthy-placenta/chorionic-villi dataset collection curated in prior work (Lykhenko et al., 2021a,b), which confined itself to Affymetrix arrays to maximise the cross-platform gene intersection. Including datasets from a wider range of vendors became practical only with imputation: without it, the intersection of all six datasets shrinks to 8,260 protein-coding genes — less than half of the ≈ 20 k human protein-coding catalog — whereas imputation recovers 17,531.

Per-dataset ENTREZID-keyed matrices were intersected with the `org.Hs.eg.db` protein-coding set; the results are summarised in Table 5. Protein-coding gene coverage ranges from 10,063 (ABI Human Genome Survey v2, GSE28551) to 18,367 (Illumina HumanHT-12 V4.0, GSE100051). This catalog-level ceiling is not reached by any single platform in our cohort — a fact frequently mis-attributed to “pipeline losses” but in reality a consequence of chip design and of updated probe reannotations, which discard probes that were originally thought to map uniquely to a single gene but no longer do.

This platform diversity creates a *block-structured* pattern of missing values: genes targeted by one platform but absent from another appear as entire rows of NAs within the corresponding

dataset blocks. Matrix completion recovers these genes by exploiting the correlation structure of the observed entries.

Dataset	Released	Platform	Tissue	Total genes	Protein-coding
GSE100051 (Soncin et al., 2018)	2018	Illumina HumanHT-12 V4.0	Placenta	19,829	18,367
GSE122214 (Zhao et al., 2019)	2018	Affy HG-U133 Plus 2.0	Chorionic villi	20,261	16,889
GSE28551 (Sitras et al., 2012)	2011	ABI Human Genome Survey v2	Chorionic villi	10,437	10,063
GSE37901 (Uuskiila et al., 2012)	2012	Affy HG-U133 Plus 2.0	Placenta	20,342	16,889
GSE93520 (Zhang et al., 2017)	2017	Agilent Whole Human Genome 4x44K	Chorionic villi	17,884	17,834
GSE9984 (Mikheev et al., 2008)	2008	Affy HG-U133 Plus 2.0	Chorionic villi	20,342	16,889

Table 5: Per-dataset ENTREZID coverage after probe-to-gene collapsing for the six datasets used in the 1.2 comparison. “Tissue” indicates whether the source article describes the sampled material as whole placenta (full-thickness biopsy) or dissected chorionic villi; justification for treating these as comparable is given in Section 5.2. “Total unique genes” counts all ENTREZIDs present in the ENTREZID-keyed TSV; “Protein-coding” is the subset with GENETYPE == "protein-coding" in org.Hs.eg.db 3.21.0.

Definition of the 1.2 contrast. Human pregnancy is conventionally divided into three trimesters: the *first trimester* (gestational weeks 1–12), which covers early placentation, trophoblast invasion, and establishment of the utero-placental circulation; the *second trimester* (weeks 13–27), during which the placenta transitions to a mature maternal–fetal exchange organ; and the *third trimester / term* (weeks 28–40, with “term” referring to samples collected at or after 37 weeks), characterised by functional maturation, preparation for parturition, and the onset of regulated senescence. Placental transcriptomic profiles differ substantially between trimesters. In this study we focus on the **1.2 comparison (First vs. Second Trimester)**, which captures the shift from early invasive placentation to the established exchange-organ phenotype. The “baseline” group is First Trimester and the “contrast” group is Second Trimester; a positive log fold-change therefore indicates a gene that is up-regulated as the placenta matures out of the first trimester. This is the same contrast used in the earlier intersection-only analyses (Lykhenko et al., 2021a,b), which makes the direct comparison of pipelines in Table 4 meaningful.

5.2 Decision to integrate placenta and chorionic villi

The six datasets in this study are labeled as either “Placenta” (full-thickness biopsy) or “Chorionic villi” (dissected villous tissue) depending on the sample preparation protocol described in each source article (Table 5). Full-thickness placental biopsies are predominantly composed of chorionic villous tissue, with minor contributions from the chorionic and basal plates and intervillous blood; explicitly dissected chorionic-villi preparations remove these peripheral layers.

In both cases the dominant cell populations — syncytiotrophoblast, cytotrophoblast, villous stroma, and fetal endothelium — are the same, and the transcriptomic difference between full-thickness and dissected-villi preparations is expected to be small relative to the gestational-age contrast under study.

We reviewed the tissue collection protocols reported in the source publications for each of the six datasets:

- **GSE100051** (Soncin et al., 2018): “random sampling of villous tissue, containing a mixed cell population, with stromal tissues as well as trophoblasts.”
- **GSE122214** (Zhao et al., 2019): “villi samples were isolated and purified immediately from the specimens according to the morphology within 30 min under an inverted microscope.”
- **GSE28551** (Sitras et al., 2012): “Chorionic tissue was dissected from a standardized location approximately 2 cm beside the umbilical cord insertion, from the middle layer of placenta midway between maternal and fetal surfaces”; first-trimester tissue “separated from decidua using light microscopy.”
- **GSE37901** (Uusküla et al., 2012): “full-thickness blocks of 1–3 cm were taken from a middle region ... involved all placental layers.”
- **GSE93520** (Zhang et al., 2017): “chorionic villus samples at 6 to 10 weeks of gestation”; quality verified by H&E staining confirming $\geq 90\%$ placental chorion.
- **GSE9984** (Mikheev et al., 2008): “the placental villus could be dissected and snap-frozen.”

All six datasets sample tissue from the villous tree of the placenta. The “Placenta” datasets (GSE100051, GSE37901) take full-thickness biopsies that are dominated by villous parenchyma, while the “Chorionic villi” datasets (GSE122214, GSE28551, GSE93520, GSE9984) explicitly dissect villous tissue from surrounding structures. No dataset in this cohort samples the basal plate, chorion laeve (chorion membrane), or decidua as its primary target — unlike GSE70102 (Bianco et al., 2016), which was excluded for this reason. GSE22490 (Rull et al., 2013) was also excluded because its tissue collection protocol describes “homogenized placental tissue (containing trophoblastic and decidual material)”; the explicit inclusion of decidual contamination makes it unsuitable for a pure villous-tissue comparison.

We consider this degree of tissue overlap sufficient for joint analysis, particularly because (a) ComBat batch correction absorbs systematic per-study offsets, including any arising from minor differences in tissue composition; (b) the gestational-age signal between first and second trimester is far larger than the expected difference between full-thickness and dissected-villi preparations of the same gestational age; and (c) the subsampling validation in Section 2.5 confirms that removing any single dataset does not qualitatively alter the differential expression result.

5.3 Pipeline overview

Figure 7 gives an end-to-end overview of the direct-merge pipeline and highlights the four configuration parameters that change the downstream behaviour most substantially: the minimum dataset-presence requirement, the per-group safety rule, the imputation method, and the imputed-cell weight used in differential expression. Each step is described in detail in the subsections that follow.

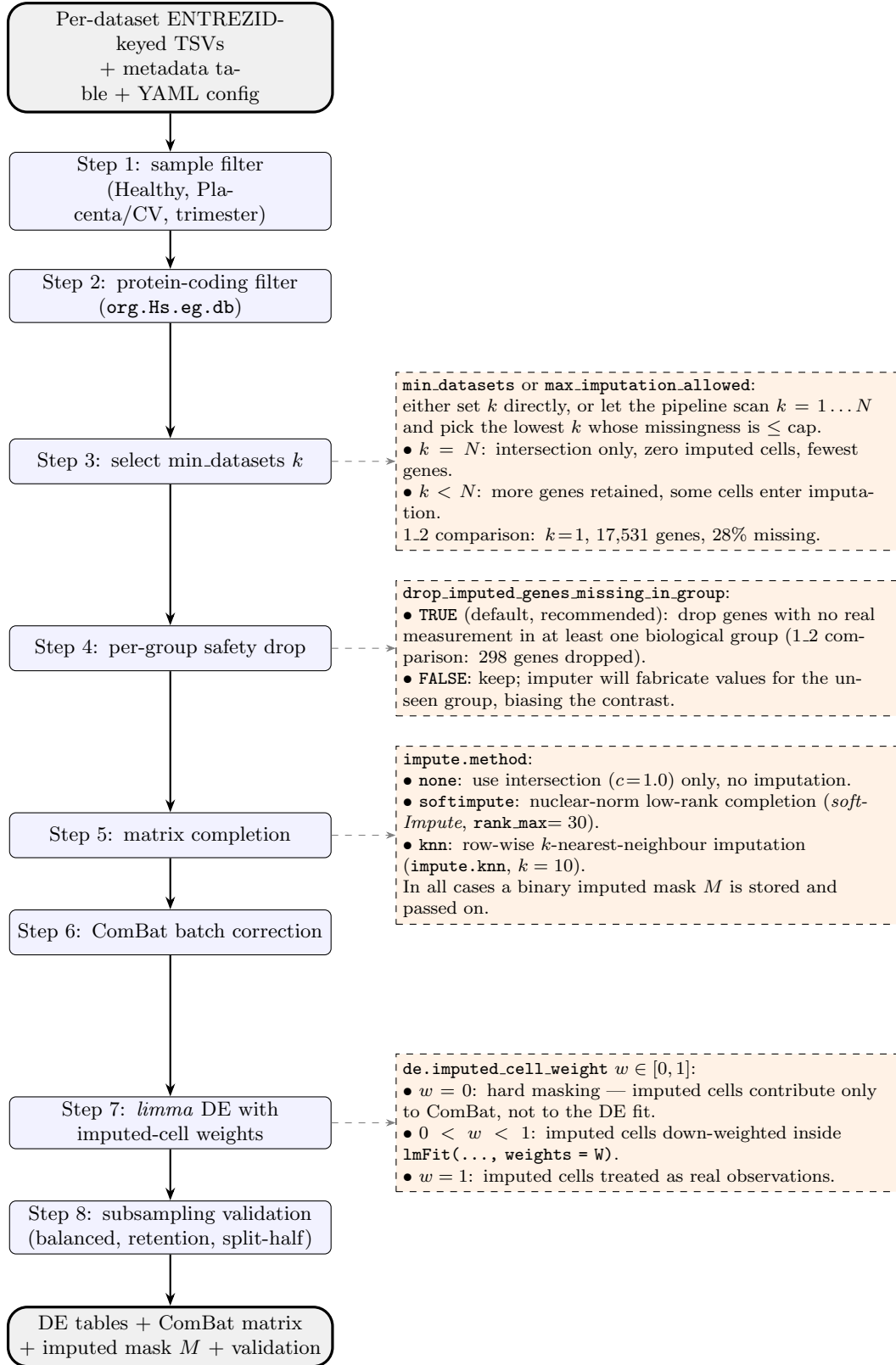


Figure 7: Direct-merge pipeline overview with parameter-dependent branching. Solid arrows show the deterministic flow from inputs (top) through the seven method steps to outputs (bottom); dashed callouts list the configurable options at each branching step and their downstream consequences. The configuration used for the results in Tables 1–4 is: `max_imputation_allowed = 1.0`, `drop_imputed_genes_missing_in_group = TRUE`, `impute.method = softimpute`, ComBat-ref (reference batch: GSE100051), and `de.imputed_cell_weight  $w = 0$` (hard masking).³³

5.4 Input harmonisation and sample filtering

The pipeline (`run_phase2b.R`³⁴³⁵) consumes a YAML configuration file that specifies (a) the directory containing ENTREZID-keyed expression TSVs, (b) the metadata table with sample-level annotations, (c) the biological contrast (baseline vs. contrast group) and any covariates (e.g. *Fetus.Sex*), and (d) tuning parameters for downstream steps. Each expression file is read and filtered to samples matching a `Diagnosis = "Healthy"`, `Biological.Specimen ∈ {"Placenta", "Chorionic villi"}`, and the two gestational-age categories of interest.

5.5 Protein-coding gene filter

This filter is applied *before* matrix completion so that matrix factorisation does not attempt to model non-coding RNAs, pseudogenes, and retired identifiers. Per-dataset expression matrices are intersected with the set of ENTREZIDs for which `org.Hs.eg.db::GENETYPE == "protein-coding"`. An audit against the current NCBI `Homo_sapiens.gene_info` release confirmed that no protein-coding genes were lost by using the installed Bioconductor snapshot.

5.6 Minimum dataset-presence selection

The user specifies the minimum number of datasets k in which a gene must be present for inclusion in the merged matrix. Two mutually exclusive configuration modes are available:

1. **Fixed min_datasets:** the value of k is set directly in the configuration file.
2. **Auto-selection via max_imputation_allowed:** the pipeline scans $k = 1, 2, \dots, N$ and selects the lowest k whose post-drop missing fraction does not exceed the cap.

5.7 Per-group imputation safety check

For every gene that requires imputation in at least one dataset, the pipeline counts the number of *non-imputed* samples available in each of the two biological groups. Genes that lack any real measurement in one of the two groups are dropped (configurable via `drop_imputed_genes_missing_in_group`), because matrix completion under a sparse row with no group-specific evidence would otherwise synthesise an arbitrary value under the unseen condition and bias the contrast.

5.8 Matrix completion

Two imputation methods are available³⁶:

1. *softImpute* (Mazumder et al., 2010; Hastie et al., 2015): low-rank matrix completion regularised by the nuclear norm (the sum of the matrix’s singular values, which penalises solutions that use more latent factors than necessary). The algorithm iteratively decomposes the observed portion of the matrix into a small number of latent factors and reconstructs the missing entries from those factors, with a warm-start λ path and column centring/scaling by `biScale`; default `rank_max = 30`, λ auto-selected along an 8-point

³⁴https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/scripts/integrative_analysis/phase2b_direct_merge/run_phase2b.R — main pipeline script

³⁵https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/scripts/integrative_analysis/phase2b_direct_merge/config_yehor_sashko/config_phase2b_1_2_yehor_6ds_no_37653_no_22490_enriched_sashko.yaml — 6ds pipeline configuration

³⁶https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/scripts/integrative_analysis/phase2b_direct_merge/imputation.R — imputation methods implementation

path. Low-rank matrix completion of this family has been used extensively for transcriptomics imputation, primarily in single-cell RNA-seq (e.g. McImpute, [Mongia et al., 2019](#)), though it is rarely applied to bulk microarray meta-analysis.

2. `impute::impute.knn` ([Troyanskaya et al., 2001](#)): k -nearest-neighbour row imputation with $k = 10$. KNN imputation is the de-facto standard for missing-value recovery in microarray studies and has been evaluated in numerous comparative reviews ([Aittokallio, 2010](#); [Liew et al., 2011](#)).

A binary `imputed_mask` is stored alongside the completed matrix and carried through all subsequent steps.

5.9 ComBat batch correction

The completed matrix is batch-corrected³⁷ with `sva::ComBat` using a model matrix of $\sim \text{bio_group} + \text{fetus_sex}$, where `bio_group` is the two-level contrast. We use the reference-batch variant of ComBat (`ref.batch` parameter), which adjusts all other batches toward the reference batch rather than toward the grand mean. GSE100051 was chosen as reference because it is the largest dataset (49 of 117 samples) and spans both comparison groups (42 first-trimester, 7 second-trimester). Parametric empirical Bayes priors are enabled.

5.10 Differential expression with imputed-cell weighting

Differential expression is computed with `limma::lmFit` followed by `eBayes` and `topTable`. A per-cell weight matrix W is constructed from the imputed mask: $W_{ij} = 1$ for real observations and $W_{ij} = w$ for imputed ones, with $w \in [0, 1]$ controlled by `de.imputed.cell.weight`. Setting $w = 0$ realises hard masking (imputed cells contribute only to batch correction, not to the DE fit), while $w = 1$ treats imputed cells as real. We use $w = 0$ (hard masking) as the default: imputed values stabilise the batch-effect estimation in ComBat by providing a complete matrix, but are excluded from the expression contrast so that differential expression calls are driven entirely by real measurements. When a dataset contains technical replicates — multiple arrays hybridised from the same biological sample — treating them as independent observations would inflate the effective sample size and produce anti-conservative p -values. The pipeline handles this via `limma`'s mixed-model machinery: a `technical_replicate_block` column in the metadata table assigns each array to its biological sample. Samples without replicates receive singleton pseudo-block identifiers. `duplicateCorrelation()` estimates the consensus intra-block correlation across all genes, and the resulting correlation and block vector are passed to `lmFit(..., block = block_vec, correlation = corfit$consensus.correlation)`, so that the linear model accounts for the within-sample covariance. This mechanism operates jointly with the imputed-cell weight matrix W when both are present. In the 1.2 comparison reported here none of the six datasets contain technical replicates, so this path is not exercised; it becomes active in other contrasts that include GSE55439 (6 biological samples profiled in 3–6 technical replicates each, 24 arrays total).

Thresholds for the significant-gene list are $\text{adj.P.Val} < 0.05$ and $|\log\text{FC}| > 1$. Differential expression is computed with `limma`'s empirical Bayes moderation ([Smyth, 2004](#); [Ritchie et al., 2015](#)).

5.11 Software and data resources

All quantitative steps are parameterised from a single YAML configuration file and logged to a timestamped `phase2b_log-<datetime>.txt`. Each run writes the imputation validation table,

³⁷https://github.com/Sashkow/integrative-gene-expression-analysis/blob/article/imputation/scripts/integrative_analysis/phase2b_direct_merge/normalization.R — batch correction implementation

the gene recovery comparison, the per-group coverage audit, the method comparison table, the per-method completed expression matrices, the per-method ComBat output matrices, and the per-method differential expression tables (both full and filtered).

The principal software dependencies are listed in Table 6.

Resource	Source	Identifier	Notes
<i>Deposited data</i>			
Raw and normalised expression data	NCBI GEO	GSE100051, GSE122214, GSE28551, GSE37901, GSE93520, GSE9984	Healthy placenta / chorionic villi
Human gene annotations	Bioconductor	org.Hs.eg.db 3.21.0	Protein-coding
<i>Software and algorithms</i>			
R (statistical environment)	R Core Team	R $\geq$ 4.3	CRAN
sva::ComBat	Leek et al.	Bioconductor	Batch correction
limma	Ritchie et al.	Bioconductor	Diff. expression
softImpute	Hastie et al.	CRAN	Matrix completion
impute::impute.knn	Troyanskaya et al.	Bioconductor	KNN imputation
AnnotationDbi	Bioconductor	—	Gene metadata
Pipeline source code	This study	[to be deposited]	Reproducible

Table 6: Software and data resources used in this study. GEO accessions list the six placental microarray datasets; gene annotations were used for protein-coding filtering. The principal R packages for batch correction, differential expression, and matrix-completion imputation are listed.

Declarations

Ethics approval and consent to participate

Not applicable. This study uses only publicly available data from the NCBI Gene Expression Omnibus (GEO). Ethics approval for each original study was obtained by the respective depositing investigators as described in the source publications.

Consent for publication

Not applicable.

Availability of data and materials

All expression datasets analysed in this study are available from the NCBI Gene Expression Omnibus under accession numbers GSE100051, GSE122214, GSE28551, GSE37901, GSE93520, and GSE9984. The pipeline source code and YAML configurations will be deposited in a public repository prior to publication.

Competing interests

The authors declare that they have no competing interests.

Funding

Authors' contributions

Acknowledgements

References

- Bianco, K., Bhatt, S., Engel, N., Glessner, J. T., & Bhatt, S. S. (2016). Placental transcriptomes in the common aneuploidies reveal critical regions on the trisomic chromosomes. *Prenatal Diagnosis*, 36(8), 760–769. doi:10.1002/pd.4862
- Aittokallio, T. (2010). Dealing with missing values in large-scale studies: microarray data imputation and beyond. *Briefings in Bioinformatics*, 11(2), 253–264. <https://doi.org/10.1093/bib/bbp059>
- Hastie, T., Mazumder, R., Lee, J. D., & Zadeh, R. (2015). Matrix completion and low-rank SVD via fast alternating least squares. *Journal of Machine Learning Research*, 16, 3367–3402.
- Hu, J., Li, H., Waterman, M. S., & Zhou, X. J. (2006). Integrative missing value estimation for microarray data. *BMC Bioinformatics*, 7, 449. <https://doi.org/10.1186/1471-2105-7-449>
- Johnson, W. E., Li, C., & Rabinovic, A. (2007). Adjusting batch effects in microarray expression data using empirical Bayes methods. *Biostatistics*, 8(1), 118–127.
- Lin, L. I-K. (1989). A concordance correlation coefficient to evaluate reproducibility. *Biometrics*, 45(1), 255–268. doi:10.2307/2532051
- Leek, J. T., Johnson, W. E., Parker, H. S., Jaffe, A. E., & Storey, J. D. (2012). The sva package for removing batch effects and other unwanted variation in high-throughput experiments. *Bioinformatics*, 28(6), 882–883.
- Li, Y., Wang, L., Zhou, J., & Ye, J. (2019). Multi-task learning based survival analysis for multi-source block-wise missing data. *Neurocomputing*, 364, 95–107. <https://doi.org/10.1016/j.neucom.2019.07.010>
- Liew, A. W.-C., Law, N.-F., & Yan, H. (2011). Missing value imputation for gene expression data: computational techniques to recover missing data from available information. *Briefings in Bioinformatics*, 12(5), 498–513. <https://doi.org/10.1093/bib/bbq080>
- Lykhenko, O., Frolova, A., & Obolenskaya, M. (2021). Consecutive integration of available microarray data for analysis of differential gene expression in human placenta. *Biotechnologia Acta*, 14(1), 38–45. <https://doi.org/10.15407/biotech14.01.38>
- Lykhenko, O., Frolova, A., & Obolenskaya, M. (2021). Changes in the human placental transcriptome during the physiological course of pregnancy. *Biopolymers and Cell*, 37(1), 73–82. <https://doi.org/10.7124/bc.000A4D>
- Mikheev, A.M., Nabekura, T., Kaddoumi, A., Bammler, T.K., Govindarajan, R., Hebert, M.F., & Unadkat, J.D. (2008). Profiling gene expression in human placentae of different gestational ages: an OPRU Network and UW SCOR Study. *Reproductive Sciences*, 15(9), 866–877. doi:10.1177/1933719108322425
- Mazumder, R., Hastie, T., & Tibshirani, R. (2010). Spectral regularization algorithms for learning large incomplete matrices. *Journal of Machine Learning Research*, 11, 2287–2322.

534 Mongia, A., Sengupta, D., & Majumdar, A. (2019). McImpute: Matrix completion based im-
535 putation for single cell RNA-seq data. *Frontiers in Genetics*, 10, 9. <https://doi.org/10.3389/fgene.2019.00009>
536

537 Rull, K., Tomberg, K., Kõks, S., Männik, J., Möls, M., Sirotkina, M., Värvi,
538 S., & Laan, M. (2013). Increased placental expression and maternal serum lev-
539 els of apoptosis-inducing *TRAIL* in recurrent miscarriage. *Placenta*, 34(2), 141–148.
540 [doi:10.1016/j.placenta.2012.11.032](https://doi.org/10.1016/j.placenta.2012.11.032)

541 Ritchie, M. E., Phipson, B., Wu, D., Hu, Y., Law, C. W., Shi, W., & Smyth, G. K. (2015).
542 limma powers differential expression analyses for RNA-sequencing and microarray studies.
543 *Nucleic Acids Research*, 43(7), e47.

544 Sitras, V., Fenton, C., Paulssen, R., Vårtun, Å., & Acharya, G. (2012). Differences in gene
545 expression between first and third trimester human placenta: a microarray study. *PLoS ONE*,
546 7(3), e33294. [doi:10.1371/journal.pone.0033294](https://doi.org/10.1371/journal.pone.0033294)

547 Soncin, F., Khater, M., To, C., Pizzo, D., Farah, O., Wakeland, A., Arul Nambi Rajan, K.,
548 Nelson, K.K., Chang, C.-W., Moretto-Zita, M., Natale, D.R., Laurent, L.C., & Parast,
549 M.M. (2018). Comparative analysis of mouse and human placentae across gestation re-
550 veals species-specific regulators of placental development. *Development*, 145(2), dev156273.
551 [doi:10.1242/dev.156273](https://doi.org/10.1242/dev.156273)

552 Smyth, G. K. (2004). Linear models and empirical Bayes methods for assessing differential
553 expression in microarray experiments. *Statistical Applications in Genetics and Molecular Bi-*
554 *ology*, 3(1), Article 3.

555 Tuikkala, J., Elo, L., Nevalainen, O. S., & Aittokallio, T. (2006). Improving missing value
556 estimation in microarray data with gene ontology. *Bioinformatics*, 22(5), 566–572. <https://doi.org/10.1093/bioinformatics/btk019>
557

558 Tuikkala, J., Elo, L. L., Nevalainen, O. S., & Aittokallio, T. (2008). Missing value imputation for
559 microarray gene expression data using histone acetylation information. *BMC Bioinformatics*,
560 9, 252. <https://doi.org/10.1186/1471-2105-9-252>

561 Uusküla, L., Männik, J., Rull, K., Minajeva, A., Kõks, S., Vaas, P., Teesalu, P., Reimand, J.,
562 & Laan, M. (2012). Mid-gestational gene expression profile in placenta and link to pregnancy
563 complications. *PLoS ONE*, 7(11), e49248. [doi:10.1371/journal.pone.0049248](https://doi.org/10.1371/journal.pone.0049248)

564 Troyanskaya, O., Cantor, M., Sherlock, G., Brown, P., Hastie, T., Tibshirani, R., Botstein, D.,
565 & Altman, R. B. (2001). Missing value estimation methods for DNA microarrays. *Bioinfor-*
566 *matics*, 17(6), 520–525.

567 Walsh, C. J., Hu, P., Batt, J., & Dos Santos, C. C. (2015). Microarray meta-analysis and cross-
568 platform normalization: integrative genomics for robust biomarker discovery. *Microarrays*,
569 4(3), 389–406. <https://doi.org/10.3390/microarrays4030389>

570 Zhang, B., Feng, L., Guo, H., Li, H., Yang, R., Wang, Y., Zhang, K., Yu, X., & Cheng, S.
571 (2017). Villi-specific gene expression reveals novel prognostic biomarkers in multiple human
572 cancers. *Journal of Cancer*, 8(14), 2793–2801. [doi:10.7150/jca.19787](https://doi.org/10.7150/jca.19787)

573 Zhao, L., Zheng, X., Liu, J., Zheng, R., Yang, R., Wang, Y., & Sun, L. (2019). The placental
574 transcriptome of the first-trimester placenta is affected by in vitro fertilization and embryo
575 transfer. *Reproductive Biology and Endocrinology*, 17, 50. [doi:10.1186/s12958-019-0494-7](https://doi.org/10.1186/s12958-019-0494-7)

576 Zheng, R., & Tang, M. (2024). Chain-linked multiple matrix integration via embedding align-
577 ment. *arXiv preprint arXiv:2412.02791*. <https://arxiv.org/abs/2412.02791>

578 Zhou, W., Han, L., & Altman, R. B. (2017). Imputing gene expression to maxi-
579 mize platform compatibility. *Bioinformatics*, 33(4), 522–528. [https://doi.org/10.1093/](https://doi.org/10.1093/bioinformatics/btw664)
580 [bioinformatics/btw664](https://doi.org/10.1093/bioinformatics/btw664)

581 Zhou, D., Cai, T., & Lu, J. (2021). Multi-source learning via completion of block-wise overlap-
582 ping noisy matrices. *arXiv preprint* arXiv:2105.10360. <https://arxiv.org/abs/2105.10360>

583 Additional files

584 Additional file 1 — Alternatives and supplements to ComBat

585 Review of batch correction alternatives (SVA, RUV, reference-batch ComBat, MNN/Harmony,
586 HarmonizR, batch-sensitized imputation, effect-size meta-analysis) for integrating microarray
587 studies with non-overlapping biological groups, with recommendations for the placenta pipeline.
588 (PDF, supplement1.pdf)

589 Additional file 2 — RUV negative control gene selection

590 Strategies for assembling negative control gene lists for RUV in placental transcriptome studies
591 across gestational ages, including empirical validation against pipeline results and a comparative
592 benchmark of RUV versus ComBat. (PDF, supplement2.pdf)